

# Final Project

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## Drive:

Link:

<https://drive.google.com/drive/folders/11Fwsf0Gh2EN77BEYg087kQquQXtwm3F?usp=sharing>

## Introduction

This project analyzes historical retail performance data from **FoodMart**, a North American retail company, covering the period from **1996 to 1998**. The dataset includes **two full years of transactional sales records**, combined with information on **customers, store locations, product performance, and returns**.

Our team conducted an in-depth analysis with the goal of understanding market trends, customer behavior, product profitability, and regional performance. Through this analysis, we identified key business challenges and opportunities. The insights derived were then used to formulate **data-driven recommendations aimed at improving operational efficiency, enhancing revenue, and optimizing regional strategies**.

## Main Figures Dashboard

### Story Explanation

This dashboard provides an overall performance overview of FoodMart, highlighting total sales, profitability, customer engagement, brand performance, and regional contribution. It sets the foundation for understanding business scale and identifying top-performing segments.

## Chart Descriptions

| Visual                                                             | Description                                                                                     |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>KPIs (Sales, Profit %, Brands, Returns, Transactions, Cost)</b> | Summarize key performance metrics to quickly evaluate business health.                          |
| <b>Top 10 Brands</b>                                               | Bar chart to show leading brands by sales volume. Helps identify high-performing product lines. |
| <b>Transactions Over Time</b>                                      | Line chart to track customer activity trends, useful for spotting seasonal patterns.            |
| <b>Member Cards Distribution</b>                                   | Pie chart to understand customer loyalty segmentation.                                          |
| <b>Top Customers Purchases</b>                                     | Horizontal bar chart to identify highest-value customers.                                       |
| <b>Sales by Region</b>                                             | Donut chart to show contribution of each region to sales performance.                           |
| <b>Top 10 Brands</b>                                               | Bar chart to show leading brands by sales volume. Helps identify high-performing product lines. |
| <b>Transactions Over Time</b>                                      | Line chart to track customer activity trends, useful for spotting seasonal patterns.            |
| <b>Member Cards Distribution</b>                                   | Pie chart to understand customer loyalty segmentation.                                          |
| <b>Top Customers Purchases</b>                                     | Horizontal bar chart to identify highest-value customers.                                       |
| <b>Sales by Region</b>                                             | Donut chart to show contribution of each region to sales performance.                           |

## Key Insights

1. The company generated **1.76M in sales with 59.67% profit**, showing strong profitability.
2. **Bronze members represent 56% of the customer base**, indicating engagement is highest at entry-level membership.
3. **North West contributes nearly half of total sales**, making it the top-performing region.

## Business Recommendations

- 1. Strengthen focus and resources on **North West and top brands (e.g., Hermanos)** to maximize profit.
- 2. Develop **loyalty upgrade programs** to move Bronze customers into higher tiers.
- 3. Analyze **returns (8,289)** to address potential product or service issues.

## Customer Insights Dashboard

### Story Explanation

This dashboard analyzes customer characteristics, purchasing behavior, and engagement trends to identify target segments and understand key demand drivers.

### Chart Descriptions

| Visual                               | Description                                                   |
|--------------------------------------|---------------------------------------------------------------|
| Total Sales vs Region                | Bar chart showing which regions generate most revenue.        |
| Customer Education Distribution      | Pie chart to segment customers by education level.            |
| Customers Over Time                  | Line chart tracking engagement trend across months.           |
| Gender vs Education                  | Column chart to reveal gender-based educational demographics. |
| Sales by Number of Children at Home  | Tree map showing family size impact on purchases.             |
| Filters: Yearly Income & Member Card | Allow deeper customer segmentation.                           |

### Key Insights

- 1. **North West leads in customer-generated revenue**, followed by Mexico Central and South West.
- 2. **~60% of customers have high school or lower education**, balanced across genders.
- 3. **Customer engagement peaks mid-year and declines in Q4**, signaling possible seasonal drop-off.

## Business Recommendations

- 1. **Run seasonal retention campaigns in Q4** to reduce decline in engagement.
- 2. Use **simple, relatable marketing tailored to high school education level customers**.
- 3. Promote **family-oriented bundles**, as households with 1–2 children show strong purchase rates.

## Product Insights Dashboard

### Story Explanation

This dashboard identifies best-selling products, profitability trends, and return patterns to optimize product strategy and supply chain decisions.

### Chart Descriptions

| Visual                       | Description                                                             |
|------------------------------|-------------------------------------------------------------------------|
| Top 50 Product Sales Table   | Highlights highest revenue products.                                    |
| Sales by Fat & Recyclability | Bar chart showing impact of health and sustainability attributes.       |
| Store State vs Returns       | Bar chart to identify regions with high return issues.                  |
| Sales & Profit by Region     | Combined bar chart to compare revenue and profitability geographically. |
| Sales & Profit Over Time     | Trend line identifying growth and fluctuations monthly.                 |
| Year Filter                  | Allows analyzing yearly performance differences.                        |

### Key Insights

- 1. **Ready-to-eat and Big Time frozen items are among the top-selling categories.**
- 2. **Recyclable, high-fat products generate higher revenue**, indicating customer preference despite health perception.
- 3. **WA has the highest return rate**, suggesting quality or service concerns.

## Business Recommendations

- 1. **Increase stock and marketing of top-selling brands (Carlson, Big Time).**
- 2. **Invest in recyclable packaging**, as it reflects higher consumer demand.
- 3. **Investigate return causes in WA & optimize store policies to reduce losses.**

## Geography Dashboard

### Story Explanation

The dashboard analyzes performance by territory, focusing on regional sales trends, product demand by region, store distribution, and return patterns to identify expansion opportunities.

### Chart Descriptions

| Visual                         | Description                                                      |
|--------------------------------|------------------------------------------------------------------|
| Top 3 Region Sales Trend       | Line chart to show regional growth pattern over time.            |
| Top 5 Product Sales in Regions | Clustered bar chart highlighting product performance per region. |
| Region vs Returns              | Bar chart to identify where product dissatisfaction is highest.  |
| Store Count by Region          | Bar chart to assess market coverage and expansion potential.     |

### Key Insights

- 1. **North West consistently outperforms other regions in sales**, while Mexico Central and South West show stable growth.
- 2. **Return rates are highest in North West**, indicating a recurring issue.
- 3. **Store density aligns with revenue, but some regions like Mexico West & Central West are underdeveloped.**

## Business Recommendations

- 1. **Expand into regions with low store presence but potential demand (Mexico West, Central West).**
- 2. **Investigate high return rate in North West**, focus on product/service improvements.
- 3. **Optimize inventory based on regional product demand forecasts.**

## Dashboard Conclusion:

The analysis of FoodMart's performance across sales figures, customer behavior, product attributes, and geographical distribution provides a comprehensive view of business health between 1996 and 1998. The company achieved strong results, generating **1.76M in sales with a 59.67% profit margin**, driven mainly by the **North West region**, which also shows the highest customer engagement and product demand. Sales are powered by **ready-made and recyclable products**, particularly from top brands like *Hermanos, Carlson, and Big Time*. Customer profiling reveals that most shoppers are **Bronze-level members with high school or lower education**, balanced across genders, and typically from **small family households**, highlighting a clear segment for targeted marketing. However, challenges persist, including **high returns (especially in North West and WA)** and **declining customer activity toward year-end**, indicating potential satisfaction and retention concerns. Additionally, some regions such as **Mexico West and Central West remain unexplored opportunities** due to low store density despite moderate demand potential. Overall, the dashboard clearly shows that **expansion in strong-performing regions, data-driven marketing tailored to customer demographics, and quality improvement initiatives will be key to sustaining growth and profitability.**